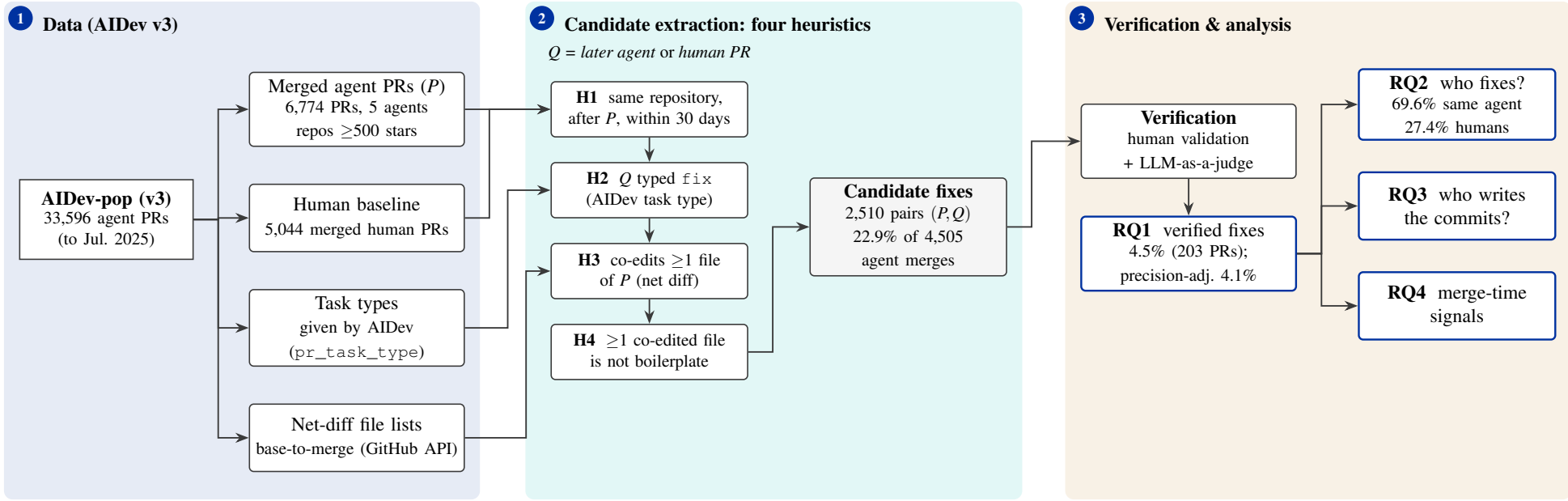

1
Data (AIDev v3)
AIDev-pop (v3)
33,596 agent PRs
(to Jul. 2025)
Merged agent PRs (P)
6,774 PRs, 5 agents
repos ≥500 stars
Human baseline
5,044 merged human PRs
Task types
given by AIDev
(pr_task_type)
Net-diff file lists
base-to-merge (GitHub API)
2
Candidate extraction: four heuristics
Q = later agent or human PR
H1 same repository,
after P, within 30 days
H2 Q typed fix
(AIDev task type)
H3 co-edits ≥1 file
of P (net diff)
H4 ≥1 co-edited file
is not boilerplate
Candidate fixes
2,510 pairs (P,Q)
22.9% of 4,505
agent merges
3
Verification & analysis
Verification
human validation
+ LLM-as-a-judge
RQ1 verified fixes
4.5% (203 PRs);
precision-adj. 4.1%
RQ2 who fixes?
69.6% same agent
27.4% humans
RQ3 who writes
the commits?
RQ4 merge-time
signals